KRITZY ESP32-S3 Robot — v5 Engineering Definition

Selected direction: 1S LiPo · 4 × SG90-class servos · plug-in peripheral modules · 2-layer PCB

Release status: PRELIMINARY ENGINEERING DEFINITION — NOT A RELEASED KICAD DESIGN. This package advances the circuit definition and load budget, but it does not claim that a native KiCad schematic, routed PCB, ERC/DRC report, or fabrication outputs have been generated or verified.

1. Power architecture

Use a protected 1-cell Li-ion/LiPo pack (nominal 3.7V, fully charged 4.2V) with two separate conversion branches: (A) a 3.3V buck-boost rail for ESP32-S3 and compatible low-power modules; (B) a dedicated high-current 5V boost rail for the four servos. Keep the servo high-current loop short and physically away from the microphone and RF antenna. Join all grounds with a deliberate low-impedance return/plane; do not route servo current through thin logic-ground traces.

Charging must use a 1-cell charger with a genuine power-path/load-sharing feature. The charger SYS output should not be assumed capable of supplying the servo rail. The servo boost is shown from the protected battery node; USB-powered simultaneous operation requires explicit input-current, thermal and cell-isolation design. Do not use a bare TP4056 board as if it automatically provides load sharing.

2. Worst-case planning current

Load / calculation	Result	Meaning
4 servos × 0.65A assumed stall	2.60A at 5V	Assumption; exact SG90 variants differ
Servo rail + 30% margin	3.38A at 5V	Minimum planning target, not final regulator selection
Servo boost input at 3.0V, 85% efficiency	6.63A from battery	Calculated: 5 × 3.38 / (3.0 × 0.85)
3.3V logic rail target	1.00A at 3.3V	ESP32 guide says supply current capacity no less than 500mA; extra ma
Logic converter input at 3.0V, 85% efficiency	1.29A from battery	Calculated: 3.3 × 1.0 / (3.0 × 0.85)
Combined estimate	7.92A from battery	Excludes startup/transients and additional audio peaks
Preliminary battery target	≥10A discharge capability candidate	Must be checked against actual cell/protection datasheets and measured

These figures are a transparent design estimate, not measured performance. The 0.65A SG90-class stall-current figure and 85% conversion efficiency are planning assumptions. A servo can briefly draw high current on startup, under load, or at mechanical stall. The exact servo datasheet, real stall current at the chosen voltage, cell continuous-discharge rating, fuse, connector and copper width must be checked before the power stages are selected.

3. Component decisions

Frozen direction: ESP32-S3-WROOM-1 module family; 1S battery; module headers for OLED, MPU6050, INMP441 and microSD; MAX98357A-compatible I2S amplifier; four 3-pin servo connectors; 2-layer PCB. Not frozen: exact ESP32 module suffix, charger/power-path IC, 3.3V regulator, high-current 5V boost, battery cell, protection/fuse, connector footprints, exact breakout board revisions, speaker impedance and mounting dimensions.

Do not select a regulator based only on its advertised switch-current limit: output current at 5V depends on battery voltage, duty cycle, efficiency, thermal limits, inductor saturation and layout. With a 1S cell, a 5V / 3.4A rail is a demanding boost-converter design.

4. Provisional GPIO map

USB D–/D+ = GPIO19/20; I2C SDA/SCL = GPIO8/9; optional MPU6050 INT = GPIO10; touch = GPIO11; microSD SPI MOSI/MISO/SCK/CS = GPIO13/12/14/21; microphone I2S BCLK/WS/SD = GPIO5/4/6; amplifier I2S BCLK/LRCLK/DIN = GPIO16/15/7; servo PWM = GPIO47/48/38/39; battery ADC = GPIO1 via a calculated divider and filter. Treat every mapping as provisional until checked against the exact module SKU, boot-strapping requirements, USB use and firmware configuration. Simultaneous I2S receive/transmit clock sharing must be verified in software.

5. Two-layer PCB layout rules

Use the top layer for components and most signals, and preserve the bottom layer as a near-continuous GND plane with minimal routing. Place the ESP32 module at the board edge with its antenna keepout clear of copper, battery, cables and metal. Place the boost inductor and switching loop close to their IC; separate them from the microphone and antenna. Place local bypass capacitors at module headers and IC supply pins; put bulk capacitance near the servo connectors. Use wide copper/pours for battery and 5V servo current, sized by the actual copper thickness, temperature rise and board fab rules. Add test points for VBAT, 5V_SERVO, 3V3, GND, EN and UART.

6. Required verification gates

• Freeze exact MPNs and footprints from manufacturer datasheets, including charger, regulators, inductor, MOSFET/diode if used, connectors and battery protection.

• Calculate converter losses and thermal rise at 3.0V input and full servo load; verify cell/protection continuous and peak current ratings.

• Verify all module headers' pin order, voltage range, onboard regulators and pull-ups; confirm microSD is 3.3V-compatible.

• Create native KiCad schematic with explicit power flags, net labels, connector pin numbers, test points and mounting holes.

• Run KiCad ERC, resolve or document every warning, then transfer to PCB and run DRC with actual footprints and board rules.

• Inspect antenna keepout, servo rail current path, return paths, USB differential routing, clearances, hole positions and connector orientation.

• Bench test current, startup dips, radio activity, audio noise and regulator/cell temperature with mechanical loads before fabrication/release.

7. Source basis and limitations

Espressif's ESP32-S3 Hardware Design Guidelines recommend a 3.3V supply with current capacity of at least 500mA, entrance decoupling, and explain that a 2-layer PCB is possible when the bottom layer remains a mostly continuous ground plane. Official reference: https://documentation.espressif.com/esp-hardware-design-guidelines/en/latest/esp32s3/index.html . Exact module and peripheral datasheets remain to be matched to the selected parts. No ERC, DRC, simulation, or physical prototype test has been performed in this package.